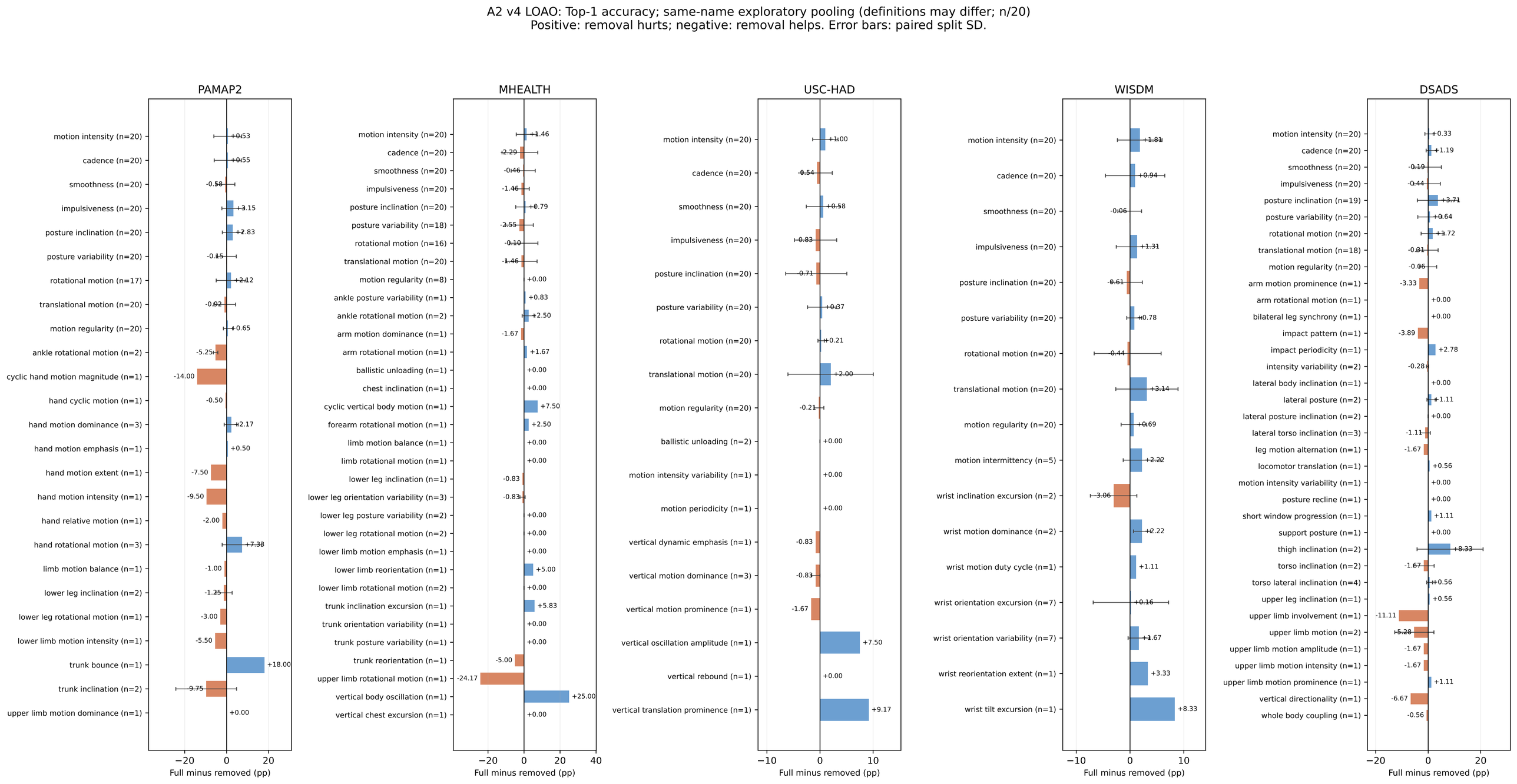
A2 v4 LOAO: Top-1 accuracy; same-name exploratory pooling (definitions may differ; n/20)
Positive: removal hurts; negative: removal helps. Error bars: paired split SD.
PAMAP2
motion intensity (n=20) +0.53
cadence (n=20) +0.55
smoothness (n=20) -0.58
impulsiveness (n=20) +3.15
posture inclination (n=20) +2.83
posture variability (n=20) -0.15
rotational motion (n=17) +2.12
translational motion (n=20) -0.92
motion regularity (n=20) +0.65
ankle rotational motion (n=2) -5.25
cyclic hand motion magnitude (n=1) -14.00
hand cyclic motion (n=1) -0.50
hand motion dominance (n=3) +2.17
hand motion emphasis (n=1) +0.50
hand motion extent (n=1) -7.50
hand motion intensity (n=1) -9.50
hand relative motion (n=1) -2.00
hand rotational motion (n=3) +7.33
limb motion balance (n=1) -1.00
lower leg inclination (n=2) -1.25
lower leg rotational motion (n=1) -3.00
lower limb motion intensity (n=1) -5.50
trunk bounce (n=1) +18.00
trunk inclination (n=2) -9.75
upper limb motion dominance (n=1) +0.00
−20 0 20
Full minus removed (pp)
MHEALTH
motion intensity (n=20) +1.46
cadence (n=20) -2.29
smoothness (n=20) -0.46
impulsiveness (n=20) -1.46
posture inclination (n=20) +0.79
posture variability (n=18) -2.55
rotational motion (n=16) -0.10
translational motion (n=20) -1.46
motion regularity (n=8) +0.00
ankle posture variability (n=1) +0.83
ankle rotational motion (n=2) +2.50
arm motion dominance (n=1) -1.67
arm rotational motion (n=1) +1.67
ballistic unloading (n=1) +0.00
chest inclination (n=1) +0.00
cyclic vertical body motion (n=1) +7.50
forearm rotational motion (n=1) +2.50
limb motion balance (n=1) +0.00
limb rotational motion (n=1) +0.00
lower leg inclination (n=1) -0.83
lower leg orientation variability (n=3) -0.83
lower leg posture variability (n=2) +0.00
lower leg rotational motion (n=2) +0.00
lower limb motion emphasis (n=1) +0.00
lower limb reorientation (n=1) +5.00
lower limb rotational motion (n=2) +0.00
trunk inclination excursion (n=1) +5.83
trunk orientation variability (n=1) +0.00
trunk posture variability (n=1) +0.00
trunk reorientation (n=1) -5.00
upper limb rotational motion (n=1) -24.17
vertical body oscillation (n=1) +25.00
vertical chest excursion (n=1) +0.00
−20 0 20 40
Full minus removed (pp)
USC-HAD
motion intensity (n=20) +1.00
cadence (n=20) -0.54
smoothness (n=20) +0.58
impulsiveness (n=20) -0.83
posture inclination (n=20) -0.71
posture variability (n=20) +0.37
rotational motion (n=20) +0.21
translational motion (n=20) +2.00
motion regularity (n=20) -0.21
ballistic unloading (n=2) +0.00
motion intensity variability (n=1) +0.00
motion periodicity (n=1) +0.00
vertical dynamic emphasis (n=1) -0.83
vertical motion dominance (n=3) -0.83
vertical motion prominence (n=1) -1.67
vertical oscillation amplitude (n=1) +7.50
vertical rebound (n=1) +0.00
vertical translation prominence (n=1) +9.17
−10 0 10
Full minus removed (pp)
WISDM
motion intensity (n=20) +1.81
cadence (n=20) +0.94
smoothness (n=20) -0.06
impulsiveness (n=20) +1.31
posture inclination (n=20) -0.61
posture variability (n=20) +0.78
rotational motion (n=20) -0.44
translational motion (n=20) +3.14
motion regularity (n=20) +0.69
motion intermittency (n=5) +2.22
wrist inclination excursion (n=2) -3.06
wrist motion dominance (n=2) +2.22
wrist motion duty cycle (n=1) +1.11
wrist orientation excursion (n=7) +0.16
wrist orientation variability (n=7) +1.67
wrist reorientation extent (n=1) +3.33
wrist tilt excursion (n=1) +8.33
−10 0 10
Full minus removed (pp)
DSADS
motion intensity (n=20) +0.33
cadence (n=20) +1.19
smoothness (n=20) -0.19
impulsiveness (n=20) -0.44
posture inclination (n=19) +3.71
posture variability (n=20) +0.64
rotational motion (n=20) +1.72
translational motion (n=18) -0.81
motion regularity (n=20) -0.06
arm motion prominence (n=1) -3.33
arm rotational motion (n=1) +0.00
bilateral leg synchrony (n=1) +0.00
impact pattern (n=1) -3.89
impact periodicity (n=1) +2.78
intensity variability (n=2) -0.28
lateral body inclination (n=1) +0.00
lateral posture (n=2) +1.11
lateral posture inclination (n=2) +0.00
lateral torso inclination (n=3) -1.11
leg motion alternation (n=1) -1.67
locomotor translation (n=1) +0.56
motion intensity variability (n=1) +0.00
posture recline (n=1) +0.00
short window progression (n=1) +1.11
support posture (n=1) +0.00
thigh inclination (n=2) +8.33
torso inclination (n=2) -1.67
torso lateral inclination (n=4) +0.56
upper leg inclination (n=1) +0.56
upper limb involvement (n=1) -11.11
upper limb motion (n=2) -5.28
upper limb motion amplitude (n=1) -1.67
upper limb motion intensity (n=1) -1.67
upper limb motion prominence (n=1) +1.11
vertical directionality (n=1) -6.67
whole body coupling (n=1) -0.56
−20 0 20
Full minus removed (pp)